ECO102: Introduction to Macroeconomics

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SECTION: 05

LECTURE 13

CHAPTER 24: FINANCIAL MARKET

The Loanable Funds Market

- The **loanable funds market** is the aggregate of all the individual financial markets.
- Funds that finance investments come from three sources:

1. Household saving, 2. Government budget surplus, 3. Borrowing from the rest of the world.

$$\mathbf{Y = C + S + T \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots 1}$$

$$\mathbf{Y = C + I + G + X - M \ldots\ldots\ldots\ldots\ldots\ldots\ldots 2}$$

$$\mathbf{I + G + X = M + S + T \quad [By\ using\ equation\ 1\ \&\ 2]}$$

$$\mathbf{Therefore,\quad I = S + (T - G) + (M - X)}$$

- This equation tells us that investment, I, is financed by household saving, S, the government budget surplus, $(T - G)$, and borrowing from the rest of the world, $(M - X)$.
- Government budget surplus $(T > G)$ contributes funds to finance investment, but a government budget deficit $(T < G)$ competes with investment for funds.

Contd.

- If we export less than we import, we borrow $(M - X)$ from the rest of the world to finance some of our investment.
- If we export more than we import, we lend $(X - M)$ to the rest of the world.
- The sum of private saving, S, and government saving, $(T - G)$, is called **national saving**
- National saving and foreign borrowing finance investment.

- The price in the loanable funds market that achieves equilibrium is an interest rate, which we also measure in real terms as the *real* interest rate.

The Real Interest Rate

- The **nominal interest rate** is the number of dollars that a borrower pays and a lender receives in interest in a year expressed as a percentage of the number of dollars borrowed and lent.
- The **real interest rate** is the nominal interest rate adjusted to remove the effects of inflation on the buying power of money.
- The real interest rate is approximately equal to the nominal interest rate minus the inflation rate.
- To see how the loanable funds market determines the real interest rate, the quantity of funds loaned, saving, and investment, ***we will ignore the government and the rest of the world and focus on households and firms in the loanable funds market.***
  - The demand for loanable funds
  - The supply of loanable funds
  - Equilibrium in the loanable funds market

The Demand for Loanable Funds

- What determines investment and the demand for loanable funds to finance it?

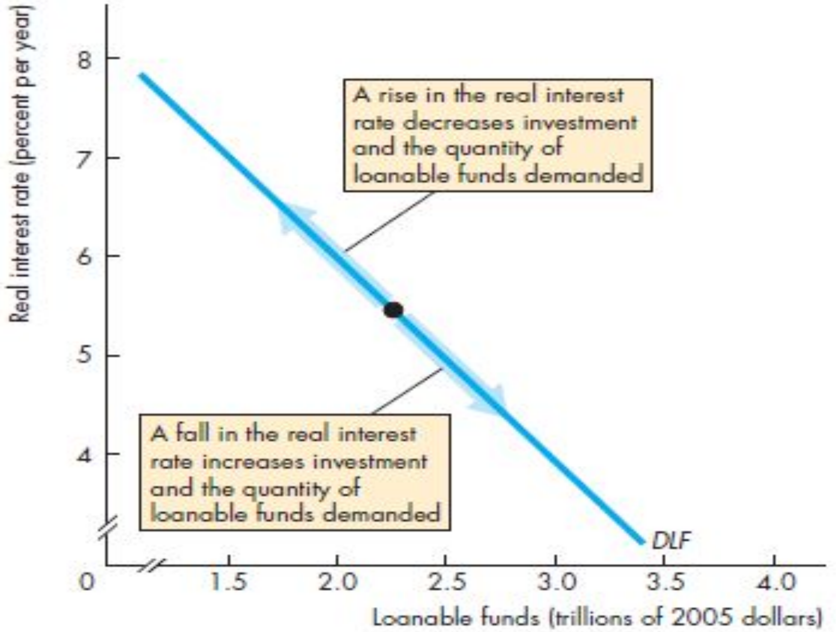
  1. The real interest rate

  2. Expected profit

- Firms invest in capital only if they expect to earn a profit and fewer projects are profitable at a high real interest rate than at a low real interest rate, so

- Other things remaining the same, the higher the real interest rate, the smaller is the quantity of loanable funds demanded; and the lower the real interest rate, the greater the quantity of loanable funds demanded.

FIGURE 24.3 The Demand for Loanable Funds



A change in the real interest rate changes the quantity of loanable funds demanded and brings a movement along the demand for loanable funds curve.

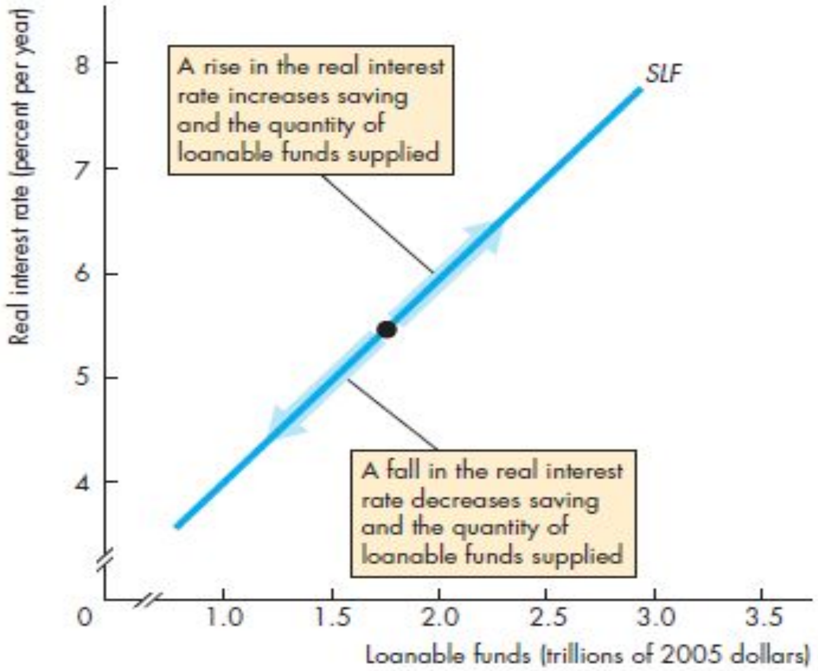
The Supply of Loanable Funds

- Your decision of supplying in loanable funds market is influenced by many factors such as-

  1. The real interest rate
  2. Disposable income
  3. Expected future income
  4. Wealth
  5. Default risk

- The **supply of loanable funds** is the relationship between the quantity of loanable funds supplied and the real interest rate when all other influences on lending plans remain the same.

FIGURE 24.4 The Supply of Loanable Funds

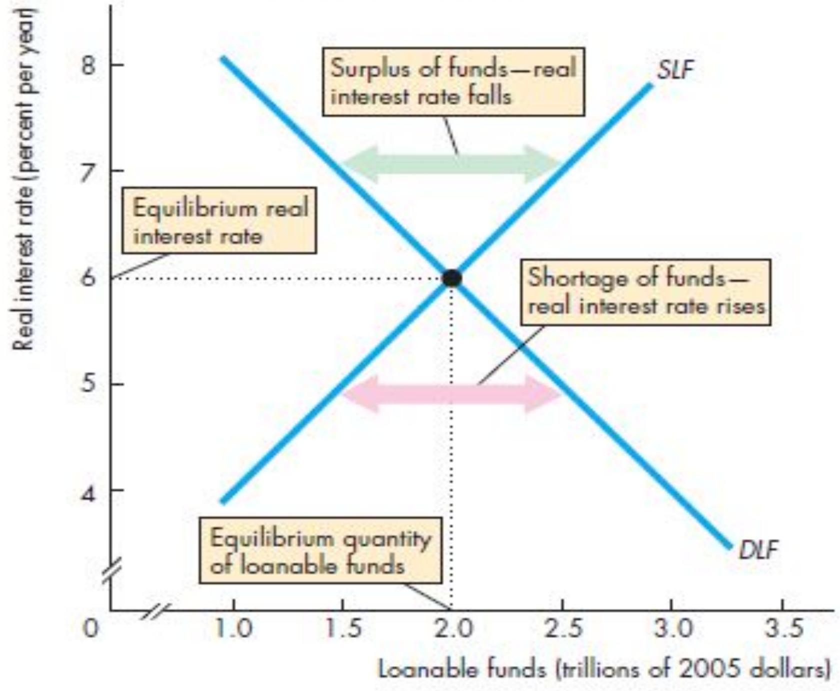


A change in the real interest rate changes the quantity of loanable funds supplied and brings a movement along the supply of loanable funds curve.

Equilibrium in the Loanable Funds Market

- There is one real interest rate at which the quantities of loanable funds demanded and supplied are equal, and that interest rate is the equilibrium real interest rate.
- Regardless of whether there is a surplus or a shortage of loanable funds, the real interest rate changes and is pulled toward an equilibrium level.
- At equilibrium interest rate, there is neither a surplus nor a shortage of loanable funds.
  - Borrowers can get the funds they want, and lenders can lend all the funds they have available.
  - The investment plans of borrowers and the saving plans of lenders are consistent with each other.

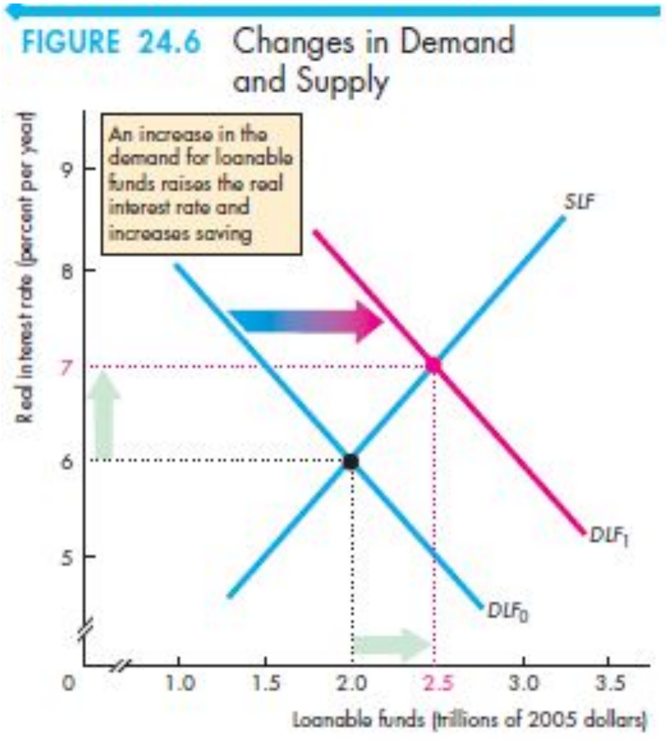
FIGURE 24.5 Equilibrium in the Loanable Funds Market



A surplus of funds lowers the real interest rate and a shortage of funds raises it. At an interest rate of 6 percent a year, the quantity of funds demanded equals the quantity supplied and the market is in equilibrium.

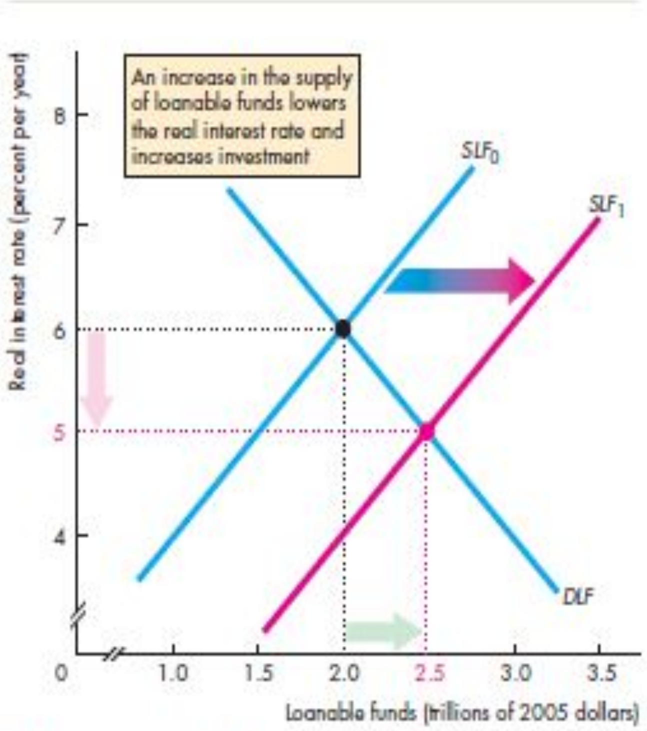
Changes in Demand and Supply

An Increase in Demand



(a) An increase in demand

An Increase in Supply

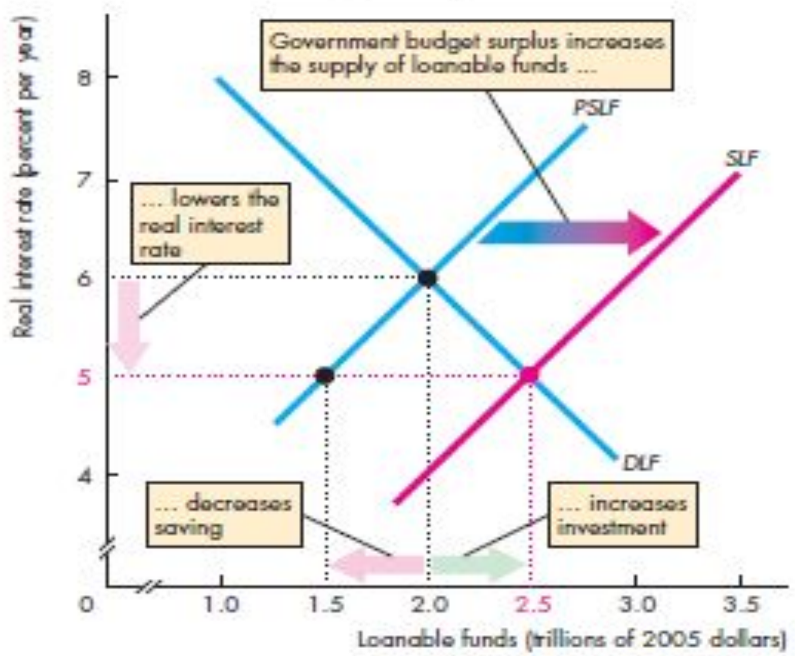


(b) An increase in supply

A Government Budget Surplus

- A government budget surplus increases the supply of loanable funds.
- The real interest rate falls, which decreases household saving and decreases the quantity of private funds supplied.
- The lower real interest rate increases the quantity of loanable funds demanded, and increases investment.

FIGURE 24.7 A Government Budget Surplus

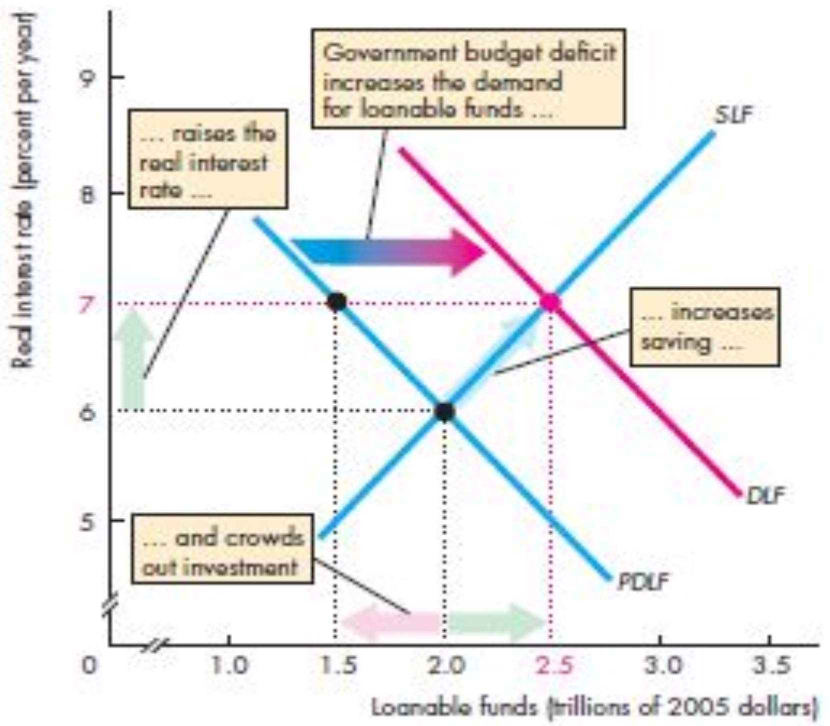


A government budget surplus of \$1 trillion is added to private saving and the private supply of loanable funds (*PSLF*) to determine the supply of loanable funds, *SLF*. The real interest rate falls to 5 percent a year, private saving decreases, but investment increases to \$2.5 trillion.

A Government Budget Deficit

- A government budget deficit increases the demand for loanable funds.
- The real interest rate rises, which increases household saving and increases the quantity of private funds supplied.
- But the higher real interest rate decreases investment and the quantity of loanable funds demanded by firms to finance investment.
- The tendency for a government budget deficit to raise the real interest rate and decrease investment is called the **crowding-out effect**.

FIGURE 24.8 A Government Budget Deficit

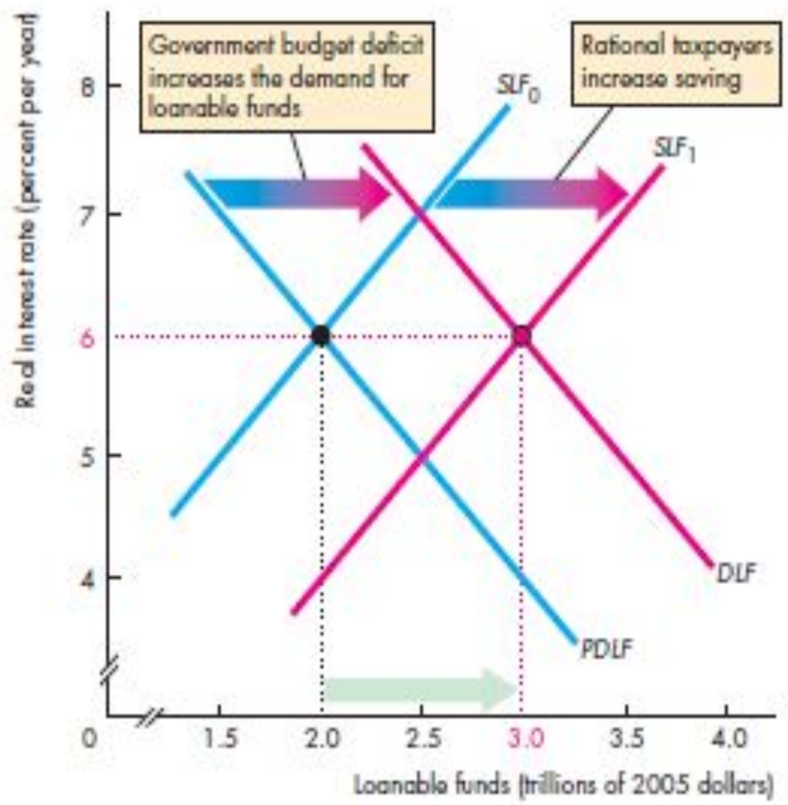


A government budget deficit adds to the private demand for loanable funds curve (*PDLF*) to determine the demand for loanable funds curve, *DLF*. The real interest rate rises, saving increases, but investment decreases—a crowding-out effect.

The Ricardo-Barro Effect

- The Ricardo-Barro effect holds that both of the effects we've just shown are wrong and the government budget, whether in surplus or deficit, has no effect on either the real interest rate or investment.
- taxpayers are rational. They can see that a budget deficit today means that future taxes will be higher and future disposable incomes will be smaller.
- With smaller expected future disposable income, savings increases today.
- Private saving and the private supply of loanable funds increase to match the quantity of loanable funds demanded by the government.
- So the budget deficit has no effect on either the real interest rate or investment.

FIGURE 24.9 The Ricardo-Barro Effect



A budget deficit increases the demand for loanable funds. Rational taxpayers increase saving, which increases the supply of loanable funds curve from SLF_0 to SLF_1. Crowding out is avoided: Increased saving finances the budget deficit.

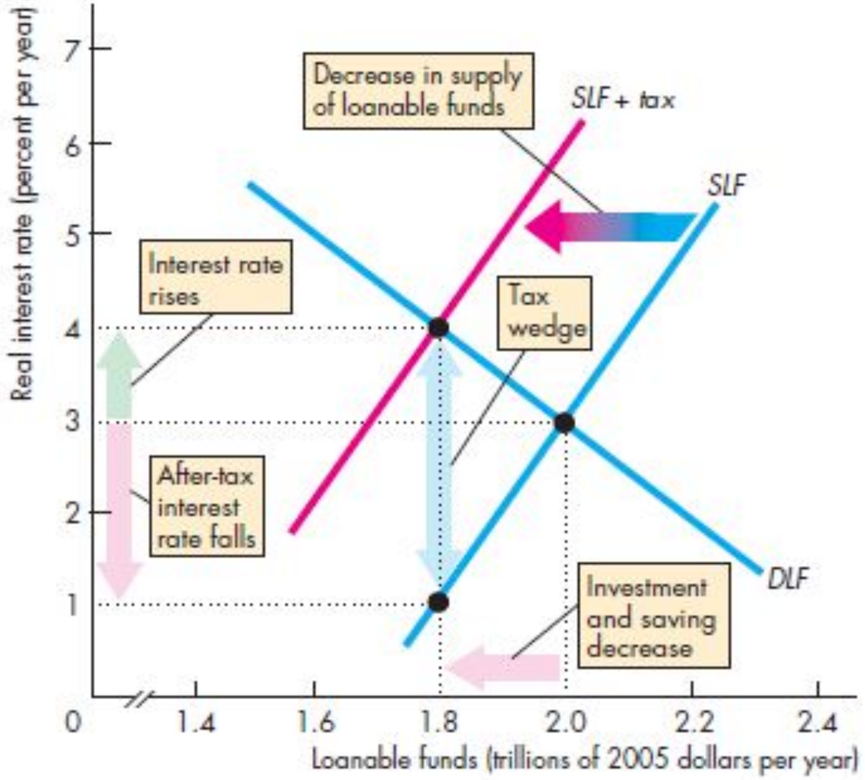
Effect of Tax Rate on Real Interest Rate

- The interest rate that influences investment and saving plans is the *real after-tax interest rate*.
- The real *after-tax* interest rate subtracts the income tax rate paid on interest income from the real interest rate.
- But the taxes depend on the nominal interest rate, not the real interest rate.
- So, the higher the inflation rate, the higher is the true tax rate on interest income.

Income Tax on Saving and Investment

- A tax on interest income has no effect on the demand for loanable funds.
- The quantity of investment and borrowing that firms plan to undertake depends only on how productive capital is and real interest rate.
- But a tax on interest income weakens the incentive to save and lend and decreases the supply of loanable funds.
- When a tax is imposed, saving decreases and the supply of loanable funds curve shifts leftward to *SLF* + *tax*.
- A tax wedge is driven between the interest rate and the after-tax interest rate, and the equilibrium quantity of loanable funds decreases.
- Saving and investment also decrease.

FIGURE 30.6 The Effects of a Tax on Capital Income

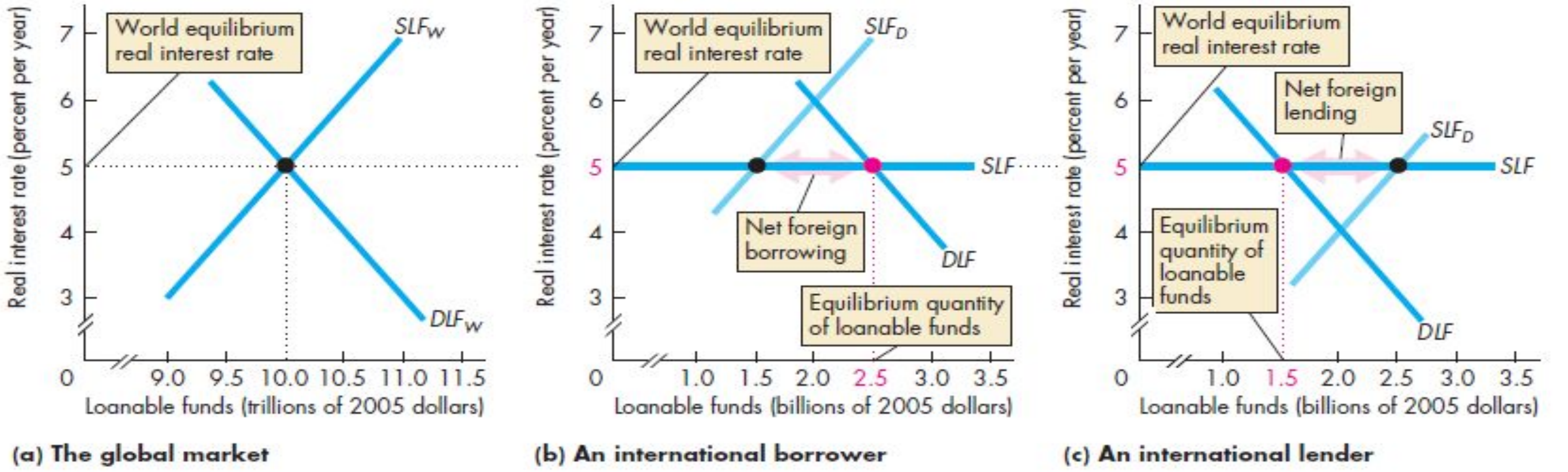


International Borrowing and Lending

- Lenders on the supply side of the market want to earn the highest possible real interest rate and they will seek it by looking everywhere in the world.
- Borrowers on the demand side of the market want to pay the lowest possible real interest rate and they will seek it by looking everywhere in the world.
- If a country's net exports are negative ($X < M$), the rest of the world supplies funds to that country and the quantity of loanable funds in that country is greater than national saving.
- If a country's net exports are positive ($X > M$), the country is a net supplier of funds to the rest of the world and the quantity of loanable funds in that country is less than national saving.

The Global Loanable Funds Market

FIGURE 24.10 Borrowing and Lending in the Global Loanable Funds Market



In the global loanable funds market in part (a), the demand for loanable funds, DLF_W, and the supply of funds, SLF_W, determine the world real interest rate. Each country can get funds at the world real interest rate and faces the (horizontal) supply curve SLF in parts (b) and (c).

At the world real interest rate, borrowers in part (b) want more funds than the quantity supplied by domestic lenders (SLF_D). The shortage is made up by international borrowing.

Domestic suppliers of funds in part (c) want to lend more than domestic borrowers demand. The excess quantity supplied goes to foreign borrowers.

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FIGURE 30.6 The Effects of a Tax on Capital Income